Greenleaf Architects Project Proposal

PROJECT LEAD: Mr THEODORE MACUPE

SERVICE PROVIDER: MACUPE TECHNOLOGIES PTY (LTD)

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About Us

Who we are

Macupe Technologies Pty (Ltd) is a well-established technology service provider that specializes in Infrastructure and Software solutions across Africa (South Africa, Kenya, Zimbabwe, Mauritius, Botswana, Namibia, Angola, D.R. Congo).

A team of passionate professionals dedicated to delivering innovative solutions that drive growth and create lasting value. At our core, we believe in integrity, collaboration, and excellence.

A well-equipped privately held company that empowers small/medium/growing organizations and individuals by providing tailored services and products that meet real-world needs while fostering long-term relationships built on trust.

Project Team

Project Lead:

Mr Theodore Macupe (A+/N+ Project Specialist)

Representing and leading our specialists

Executive Summary

Purpose

The purpose of this solution is to supply, configure, and validate a **secure small-office/home-office (SOHO) network** for Greenleaf Architects, that:

- *Segregates staff and guest traffic,*
- *Centralizes files and backups,*
- *Demonstrates strong security controls, recoverability, and auditability,*
- *Is cost-effective, scalable, and easy to operate by internal staff.*

Our Offer at a Glance

Our offer covers key infrastructure components and a cost-effective budget, highlighted below:

- ***Architecture:*** *Two logically isolated networks—**StaffNet (10.10.10.0/24)** and **GuestNet (10.10.20.0/24)**—with controlled access to central file and backup services.*

- ***Security Controls:*** *Host-level firewall policies enforcing allow-staff / block-guest for SMB (445/139), RDP (3389), and ICMP; least-privilege shares; change control and evidence capture.*

- ***Data Protection:*** *Central **NAS with RAID1**, daily incremental and weekly full backups, snapshot capability, and defined retention (**30 daily / 4 weekly / 3 monthly**).*

- ***Validation:*** *Repeatable test plan with screenshots and logs showing staff access works, guest access is denied, and backups are running and restorable.*

- ***Value for Money:*** *Business-grade hardware sized for government SOHO needs with a conservative **hardware estimate ≈ ZAR 20,850**, well within typical budget envelopes, and with room for growth.*

Executive Summary (Ctd.)

Compliance & Governance

We commit to compliance with applicable public-sector requirements, including:

- **POPIA** (security safeguards and processing limitations),
- **PFMA/MFMA** value-for-money principles (fit-for-purpose, economical procurement),
- Government **SCM** processes (competitive sourcing, record-keeping),
- Alignment to **ISO/IEC 27001** control intent (access control, backups, logging) for the scope of this engagement.
- All configurations, changes, and tests are documented for audit. Evidence packs (screenshots, rule exports, backup job reports) are delivered with the as-built.

Executive Summary (Ctd)

Deliverables

As-Built Documentation: Topology, IP addressing, firewall rule sets, share and permission matrix.

Evidence Pack: Test screenshots, firewall exports, backup schedules and last-run status, restore proof.

Operations Runbook: Backup/restore steps, routine checks, change procedure, incident quick-reference.

Asset & Configuration Register: Hardware/VM inventory, versions, license keys (if applicable).

Training & Handover: 1–2 hour administrator handover with Q&A and sign-off template.

Executive Summary (Ctd.)

Benefits to Greenleaf Architects

Security & Compliance: Clear staff/guest separation, documented controls, and auditable evidence.

Continuity: RAID1 + snapshots + scheduled backups improve recovery time and recovery point objectives.

Cost-Effectiveness: Business-grade, readily available hardware; right-sized to need; low operational overhead.

Skills Transfer: Simple, documented procedures enable internal teams to operate and support the environment.

3. Network Design and Planning

Network Design Response Plan

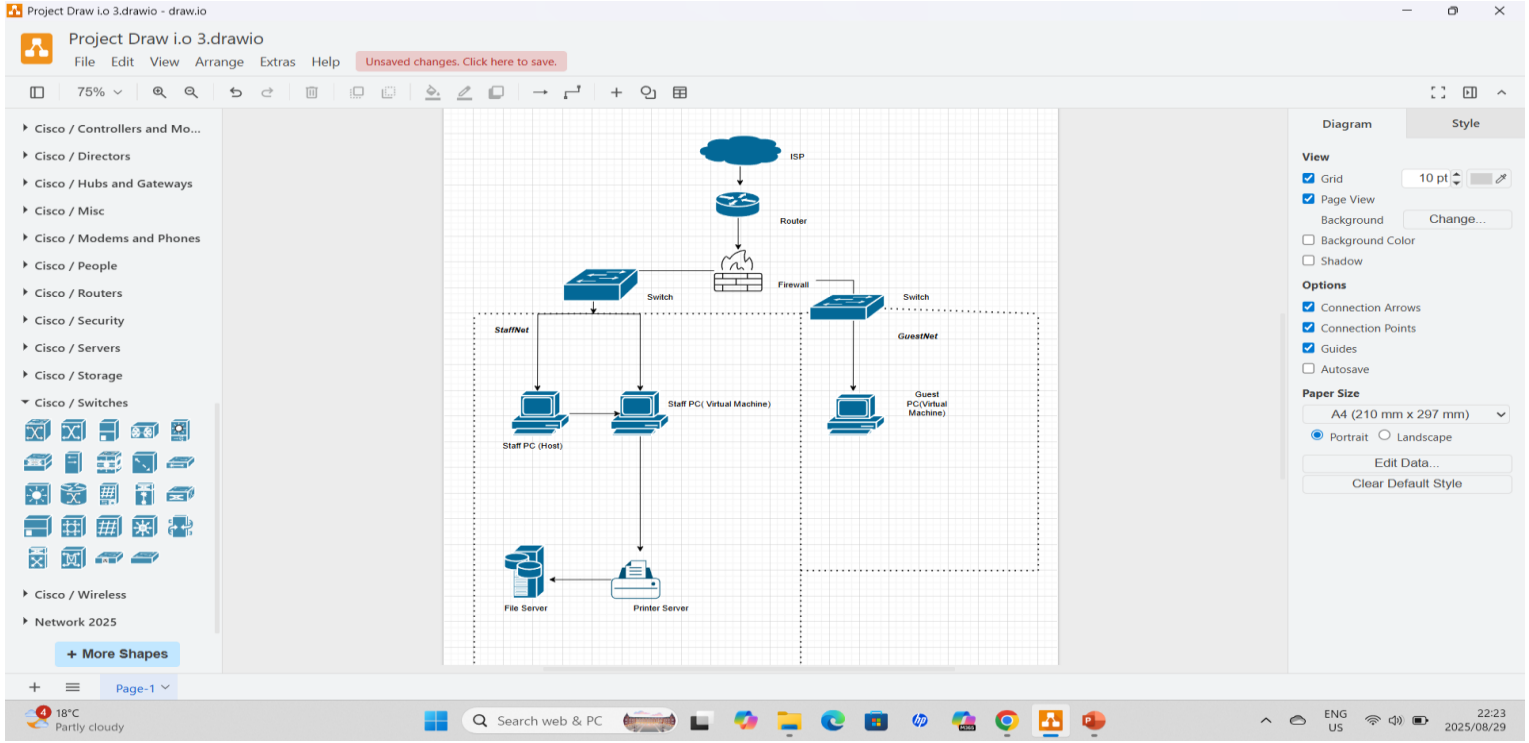
3.1. Network Topology

3.2. Budget Sheet

3.3. Virtualization

3.4. Response Summary

3.1 Network Topology



Pros

Allows for scalability

Cost Effective only 2 Switches

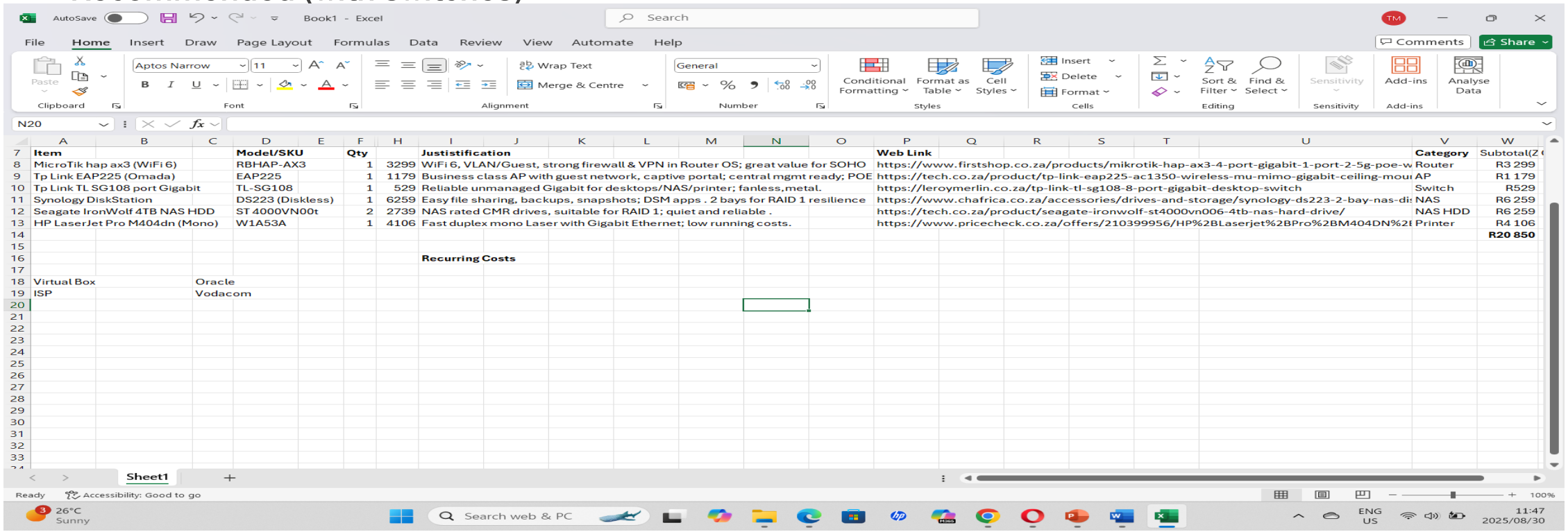
Cons

2 switches may result in congested network traffic

Screenshot Courtesy of Draw io

3. Budget Sheet (Ltd)

* Recommended (with switches)



Item	Model/SKU	Qty		Justistification	Web Link	Category	Subtotal(Z
MicroTik hap ax3 (WiFi 6)	RBHAP-AX3	1	3299	WiFi 6, VLAN/Guest, strong firewall & VPN in Router OS; great value for SOHO	https://www.firstshop.co.za/products/mikrotik-hap-ax3-4-port-gigabit-1-port-2-5g-poe-w	Router	R3 299
Tp Link EAP225 (Omada)	EAP225	1	1179	Business class AP with guest network, captive portal; central mgmt ready; POE	https://tech.co.za/product/tp-link-eap225-ac1350-wireless-mu-mimo-gigabit-ceiling-mou	AP	R1 179
Tp Link TL SG108 port Gigabit	TL-SG108	1	529	Reliable unmanaged Gigabit for desktops/NAS/printer; fanless,metal.	https://leroymerlin.co.za/tp-link-tl-sg108-8-port-gigabit-desktop-switch	Switch	R529
Synology DiskStation	DS223 (Diskless)	1	6259	Easy file sharing, backups, snapshots; DSM apps . 2 bays for RAID 1 resilience	https://www.chafrica.co.za/accessories/drives-and-storage/synology-ds223-2-bay-nas-di	NAS	R6 259
Seagate IronWolf 4TB NAS HDD	ST 4000VN00t	2	2739	NAS rated CMR drives, suitable for RAID 1; quiet and reliable .	https://tech.co.za/product/seagate-ironwolf-st4000vn006-4tb-nas-hard-drive/	NAS HDD	R6 259
HP LaserJet Pro M404dn (Mono)	W1A53A	1	4106	Fast duplex mono Laser with Gigabit Ethernet; low running costs.	https://www.pricecheck.co.za/offers/210399956/HP%2BLaserjet%2BPro%2BM404DN%2	Printer	R4 106
							R20 850
				Recurring Costs			
Virtual Box	Oracle						
ISP	Vodacom						

Screenshots Courtesy of Microsoft Excel

3.3 Virtualization

Goal: The goal with virtualization is to ensure abstraction, agility, resilience, security and scalability are achieved. This is to improve utilization and create support efficiency, while isolating safely.

Key Concepts:

* **Cloning/Replication:** Setup/Create a “gold” Staff VM → **linked clone** to Staff-PC2 to save disk/time.
* **Sandboxing:** Guest-PC on **GuestNet** preventing accidental cross-access; NAT isolates from the host LAN.
* **VRT Execution:**

	Testing	Security	Disaster Recovery (DR)
Benefits	Isolation, Repeatability, Cost & Speed Assurance, Practice DR	Strong isolation and smaller blast radius, Safe sandboxes for risky tasks, Stronger data protection features, BC/DR advantages	Lower RPO/RTO, Portability, Fast-consistent recovery, Application-consistent backups, Security & isolation, version-driven recovery points
Achieved By	Reproduction Time vs Response time tracking, Parallel testing, Network Isolation	Use NAT for internet and no default gateway, Disable unneeded virtual hardware, Turn on Secure boot, Block SMB, RDP and ICMP	Non-disruptive DR testing, Perform a cold standby, Separate backups media, Run DR drill (disconnect PS import standby, reattach shares, validate client access

3.4 Response Summary

For Greenleaf Architects, we've built a secure, simple SOHO network that just works. Staff and guests are separated into two virtual networks— **StaffNet** and **GuestNet**— so client files stay private, while guests still have internet.

A central **Synology NAS** gives us fast file sharing and **automated daily backups** with snapshots, so we can recover files in minutes. With NAS, we can have a File/Backup Server (Primary) and a NAS Server (secondary).

The proposal covers one budget sheet, with a low-cost estimates recommended (for short term, requiring an early upgrade) that according to our judgement will still be **cost-effective** and business-grade, keeping the **total under R70,000**.

The design is **scalable**: we can add users, expand storage, or adopt cloud backup without redesigning. The outcome is a reliable, secure office network that supports your growth with minimal fuss.

4. VM Simulation

4.1. VM Configuration

4.2. VLAN via Firewall/Internal Network

4.3. Network Isolation (StaffNet vs GuestNet)

4.4. Data Backup Strategy

4.5. Windows Firewall Security Rules (ISO based)

4.6. Some Core Functionalities Demonstrated

4.1 VMs Configuration

- **VM Management Tool:** Virtual Box – an Oracle VMware. Version: 7.2.0 or above, for Windows.
- **Virtual Machines Creation:**

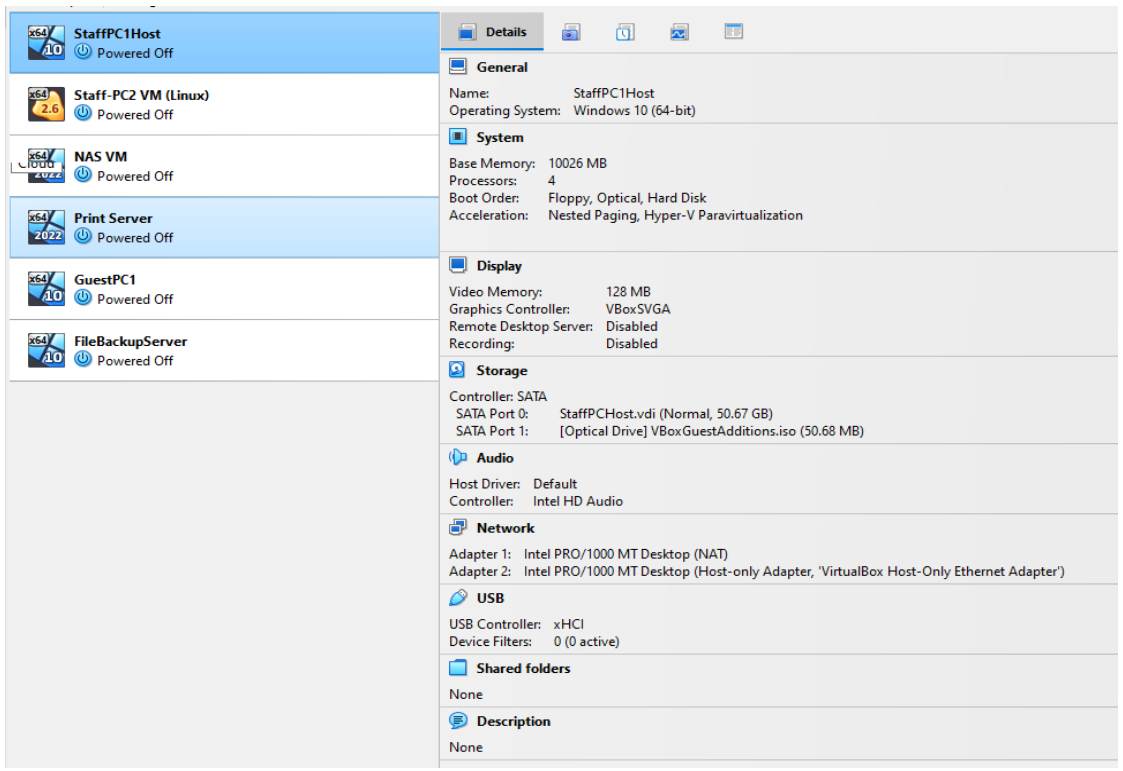
	VM1	VM2	VM3	VM4	VM5
Staff Network	Staff-PC1 Host (Win10)	Staff-PC2 (Linux)	File/Backup Server (WinServ22)	NAS (WinServ22)	Printer (WinServ22)
Guest Network	Guest-PC1 (Win10)				

- **Network Management:** Virtual Box Host-Only Ethernet Adapter (vbxnet0). This Adapter draws Internet from the staff NIC (i.e., NIC1 – StaffNet).

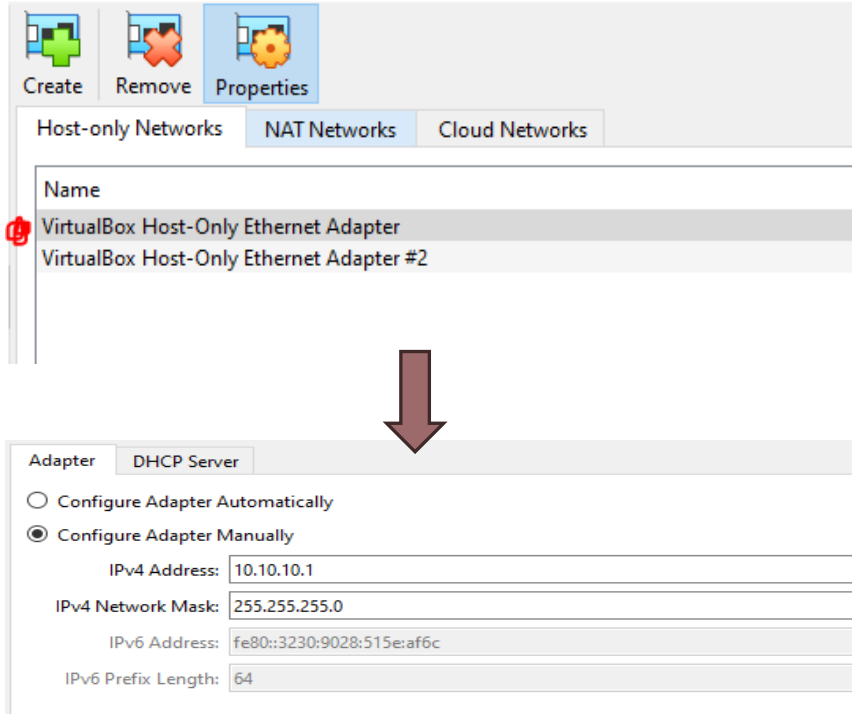
4.1 VMs Configuration (Ltd.)

* VM Creation and Configuration

VMs



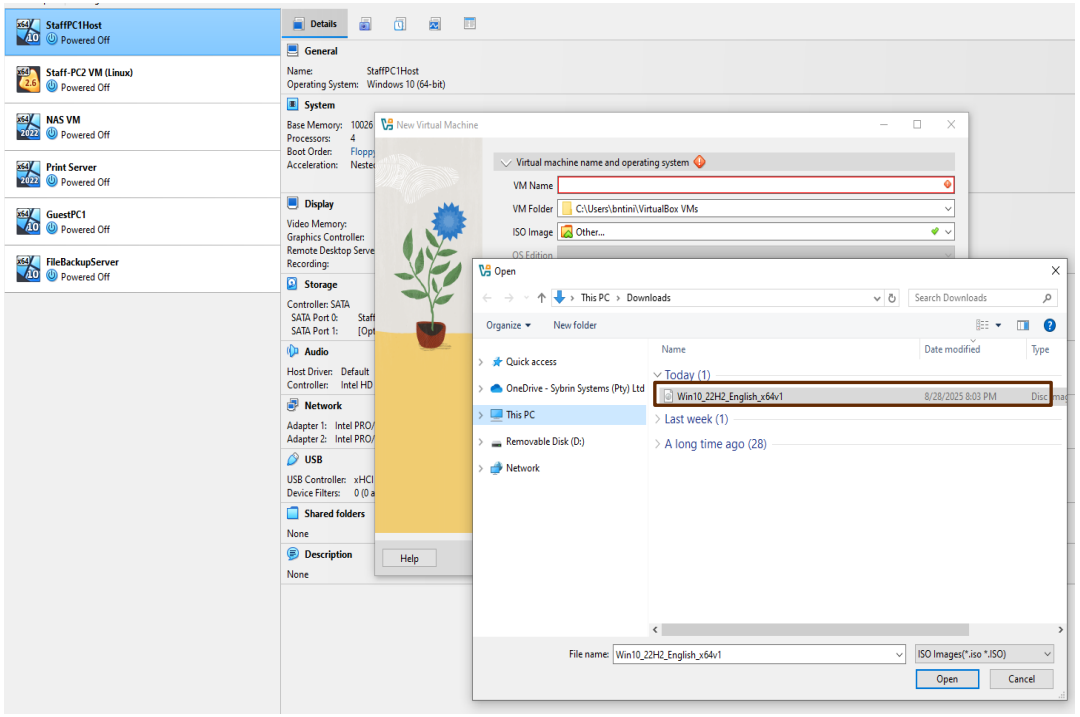
* Network Configuration

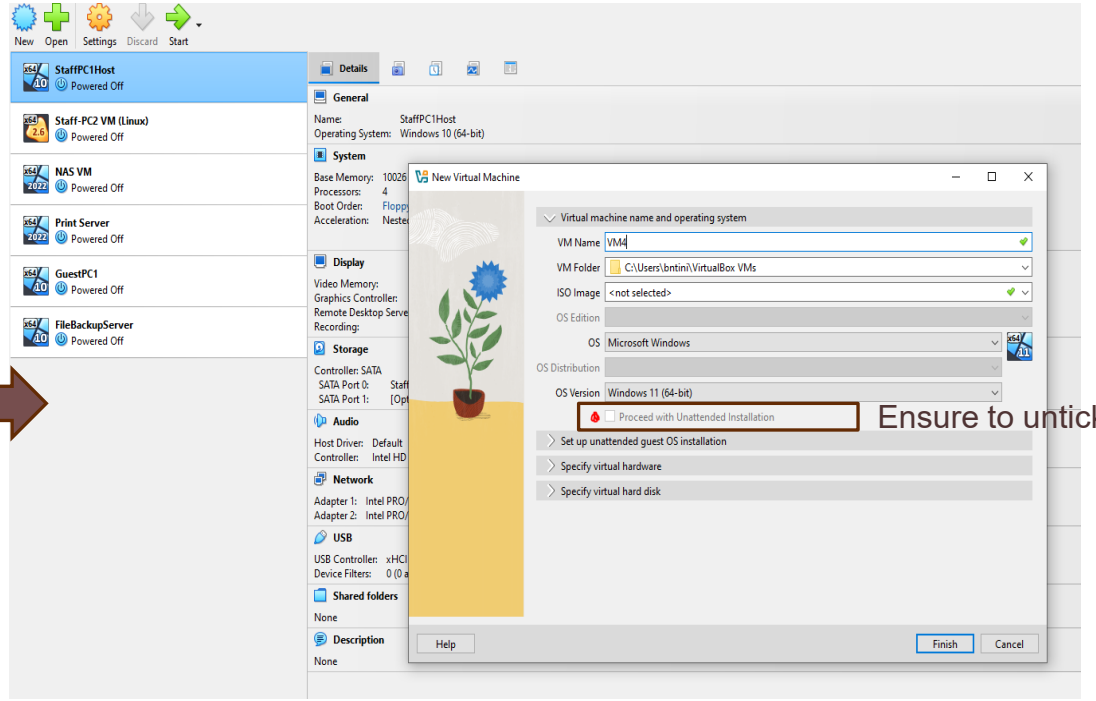


Screenshot Courtesy of Oracle Virtual Box

4.1 VMs Configuration (Ctd.)

* Setup a VM with Window 10 ISO Image



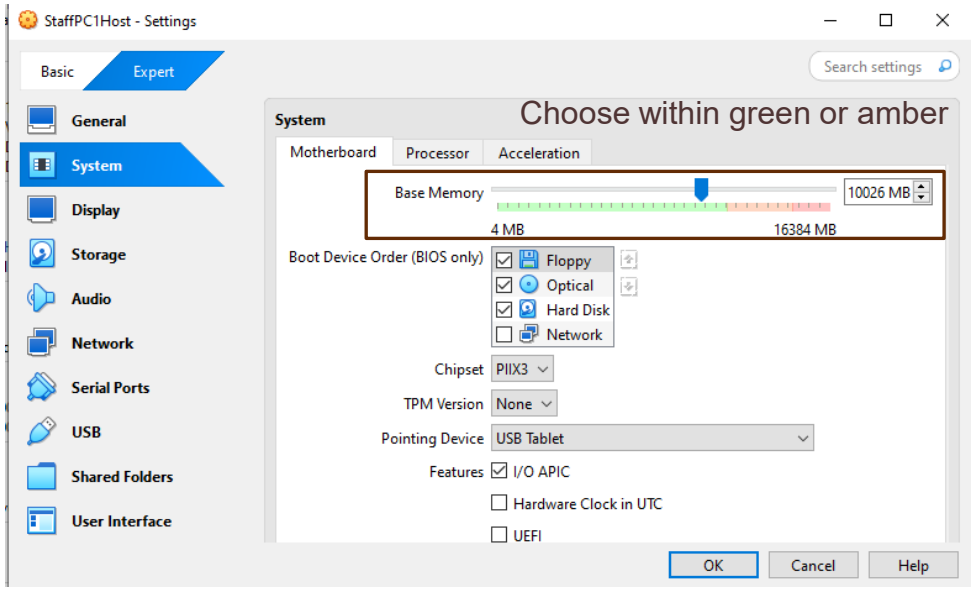


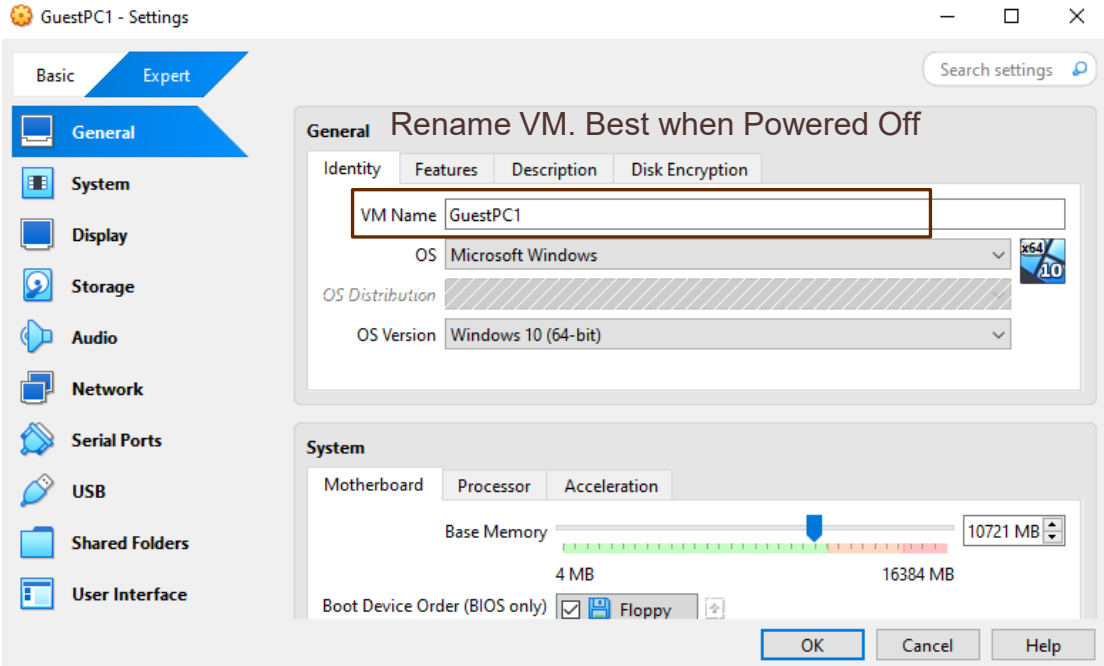
Ensure to untick

Screenshot Courtesy of Oracle VirtualBox

4.1 VMs Configuration (Ltd.)

* Setup VM RAM, CPU, HDD sizes

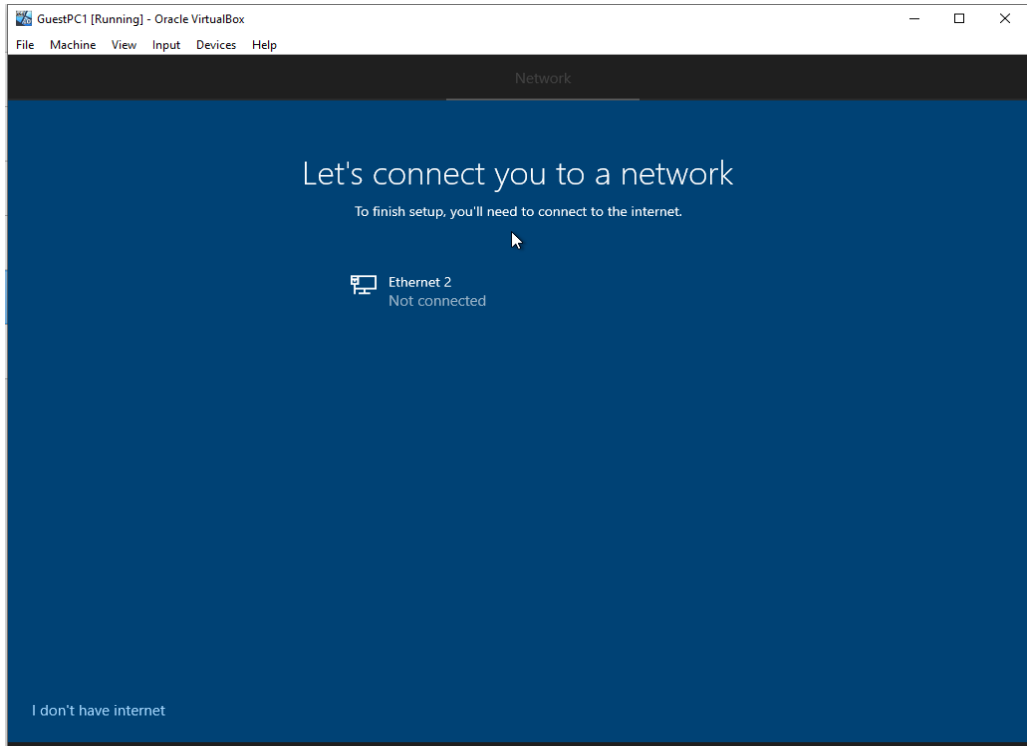


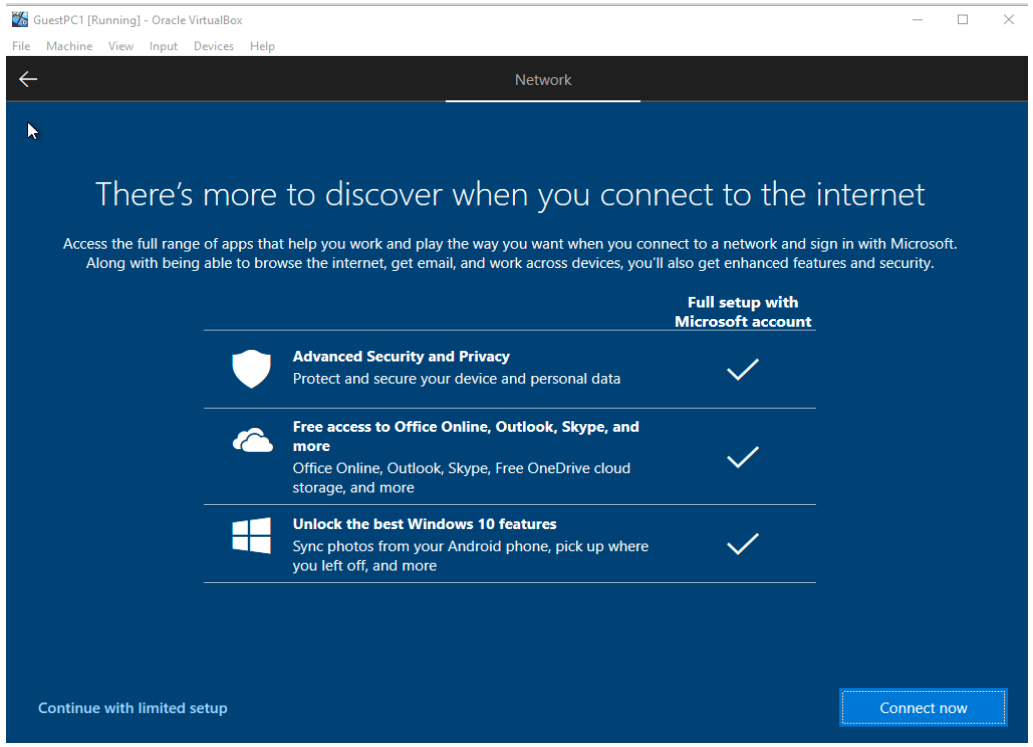


Screenshot Courtesy of Oracle VirtualBox

4.1 VMs Configuration (Ctd.)

* Complete Setup of a VM on Guest PC (Power On after setting RAM, Disk, HDD)





Screenshots Courtesy of Windows

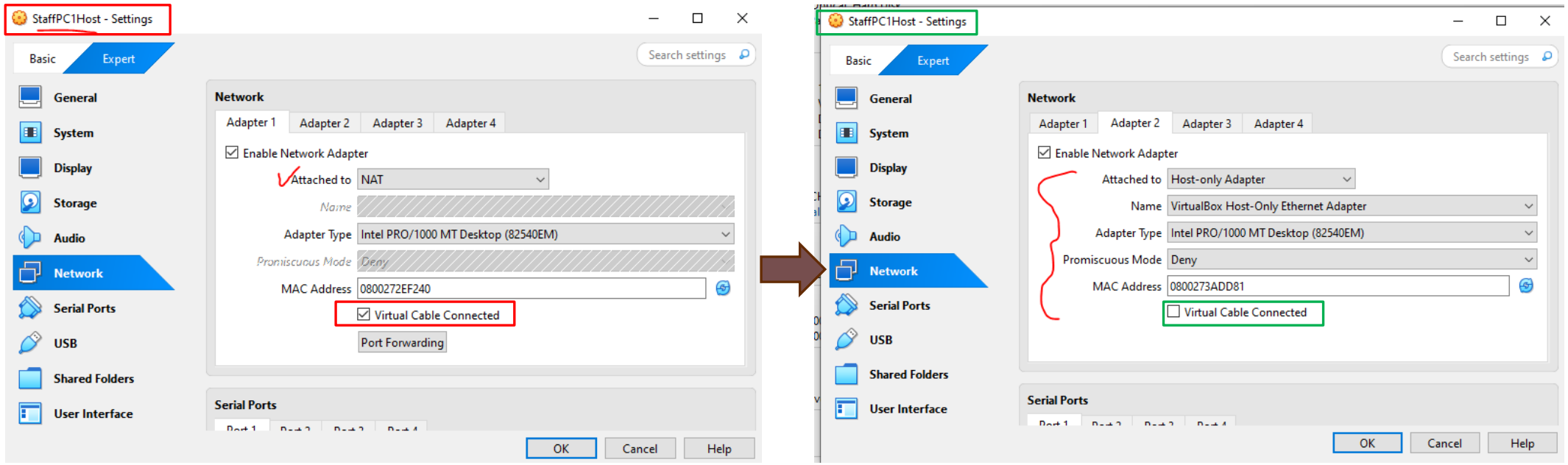
4.2 VLAN via Firewall/Internal Network

- Separation of networks using internal/Host-only adapters:
  Why it matters? This mirrors **VLAN ACLs** at host level—guests can't probe staff assets, but staff still work normally. NAT keeps "internet" outside of this policy.

	VLAN	Adapter 1	Adapter 2	Adapter Type	Virtual Cable Connected	IP
Staff-PC1 Host (Win10)	StaffNet (Vbxnet0)	NAT	Host-only	Intel PROD/1000 MT Desktop	Yes (on NAT)	10.10.10.100
Staff-PC2 (Linux)	StaffNet (Vbxnet0)	NAT	Host-only	Intel PROD/1000 MT Desktop	No	10.10.10.12
NAS (WinServ22)	StaffNet (Vbxnet0)	NAT	Host-only	Intel PROD/1000 MT Desktop	No	10.10.10.14
File/Backup Server (WinServ22)	StaffNet (Vbxnet0)	NAT	Host-only	Intel PROD/1000 MT Desktop	No	10.10.10.10
Guest-PC1 (Win10)	GuestNet	NAT	Internal Network "GuestNet"	Intel PROD/1000 MT Desktop	No	10.10.20.11

4.2 VLAN via Firewall/Internal Network (Ltd.)

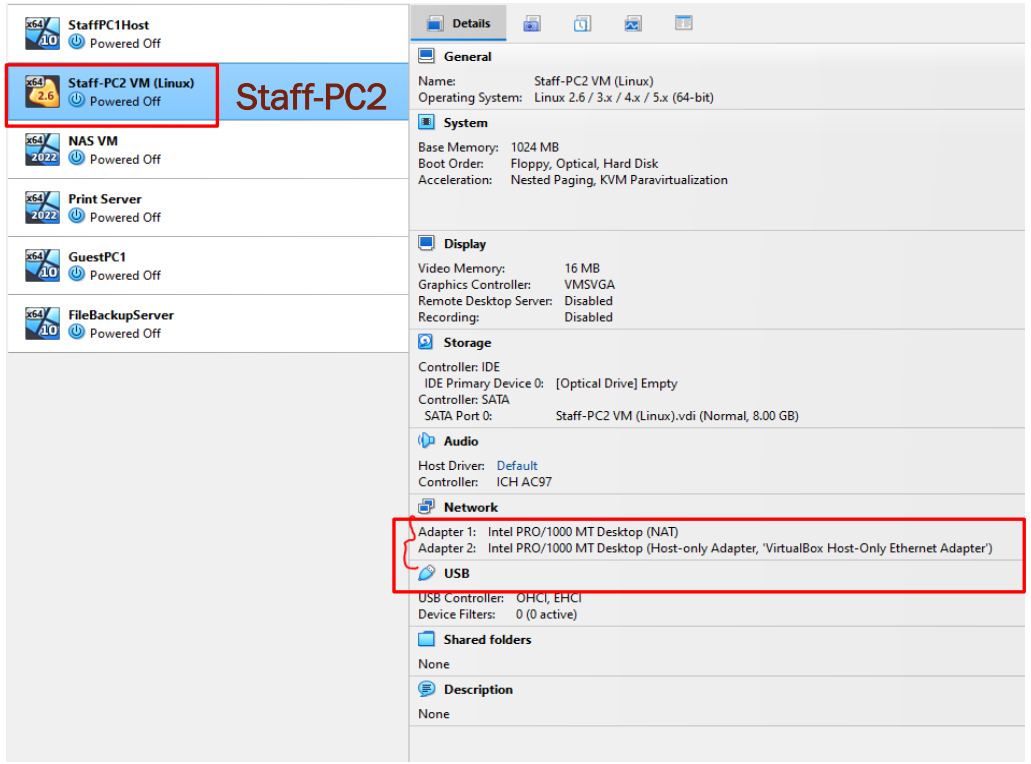
<u>* Illustration (1) of VM's Network Adapter Configuration</u>



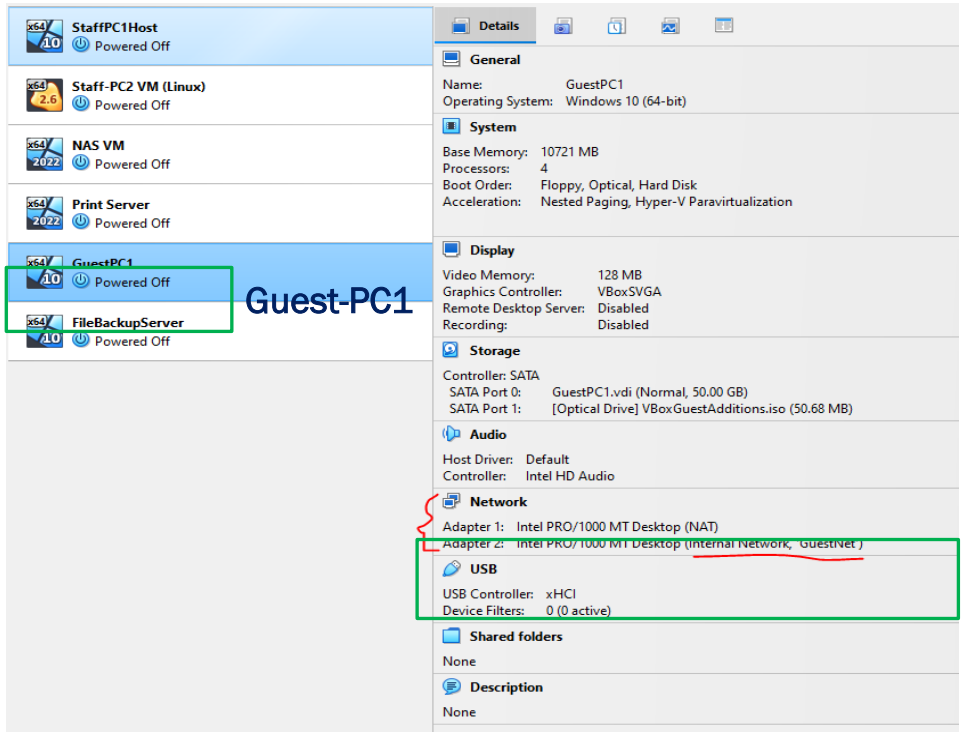
Screenshot Courtesy of Oracle VirtualBox

4.2 VLAN via Firewall/Internal Network (Ctd.)

*** Illustration (2) of Staff-PC network setup vs Guest-PC**



VS



Screenshot Courtesy of Oracle VirtualBox

4.2 VLAN via Firewall/Internal Network (Ltd.)

- Block Guest-PC from accessing staff shared resources (e.g. File Server).
  Why it matters? This is to ensure that when testing resource access, guest cannot access staff resources.

  <u>* Illustration (3) – File/Backup Server firewall rules:</u>
  Set to block any inbound request from guests. Despite network segregation, this will ensure that no bypass is achieved in the event of network bridge.

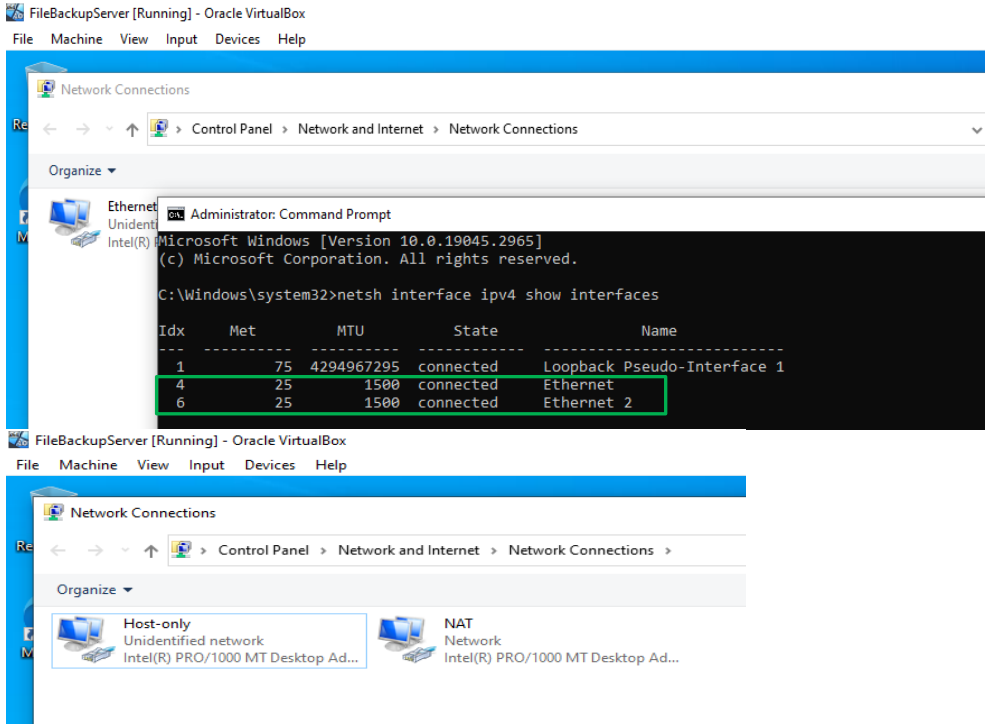
Block ICMP from GuestNet	All	Yes	Block	No	Any	Any	10.10.20.0/24	ICMPv4	Any	Any	Any	Any
Block RDP from GuestNet	All	Yes	Block	No	Any	Any	10.10.20.0/24	TCP	3389	Any	Any	Any
Block SMB from GuestNet	All	Yes	Block	No	Any	Any	10.10.20.0/24	TCP	445, 139	Any	Any	Any

Screenshot Courtesy of Windows Firewall

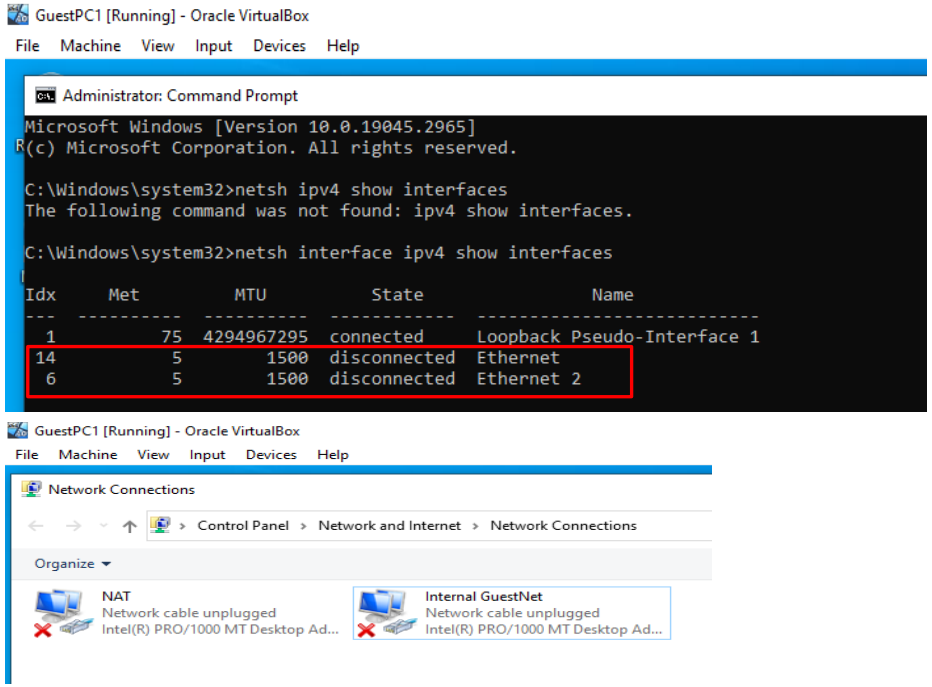
4.2 VLAN via Firewall/Internal Network (Ctd.)

- Simulated internet access

*** Illustration (4) – View of Network Access Staff VLAN**



*** Illustration (5) – View of Stopped Network Access Guest VLAN**



Screenshots Courtesy of Windows Command Prompt

4.2 VLAN via Firewall/Internal Network (Ctd.)

- Some evidence/illustrations of firewall rule configurations

*** Illustration (6) – Inbound Staff Allow Rule (File/Backup Server)**

Administrator: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> # Allow StaffNet
PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Allow SMB from StaffNet" -Direction Inbound -Action Allow -Protocol TCP -LocalPort 445,139 -RemoteAddress 10.10.10.0/24


Name                          : {d28b0c65-7a41-44fa-b555-c5b026ce6f0d}
DisplayName                   : Allow SMB from StaffNet
Description                   :
DisplayGroup                  :
Group                         :
Enabled                       : True
Profile                       : Any
Platform                      : {}
Direction                     : Inbound
Action                        : Allow
EdgeTraversalPolicy           : Block
LooseSourceMapping            : False
LocalOnlyMapping              : False
Owner                         :
PrimaryStatus                 : OK
Status                        : The rule was parsed successfully from the store. (65536)
EnforcementStatus             : NotApplicable
PolicyStoreSource             : PersistentStore
PolicyStoreSourceType         : Local
RemoteDynamicKeywordAddresses :
PolicyAppId                   :
```

PowerShell "Allow" Configuration

Allow SMB
Allow RDP
Allow ICMP

Windows Defender Firewall with Advanced Security

File Action View Help

Windows Defender Firewall with Inbound Rules
- Inbound Rules
- Outbound Rules
- Connection Security Rules
- Monitoring

Inbound Rules

Name	Group	Profile	Enabled	Action	Override	Program	Local Address	Remote Address	Protocol	Local Port	Remote Port	Authorized Users	Authorized Compute
Allow ICMP from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	ICMPv4	Any	Any	Any	Any
Allow RDP from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	TCP	3389	Any	Any	Any
Allow SMB from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	TCP	445, 139	Any	Any	Any

*** Illustration (7) – Inbound Guest Block Rule (File/Backup Server)**

```
PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Block SMB from StaffNet" -Direction Inbound -Action Block -Protocol TCP -LocalPort 445,139 -RemoteAddress 10.10.10.0/24


Name                          : {70882b81-ed3b-4bd0-93e0-e5b07d20534c}
DisplayName                   : Block SMB from StaffNet
Description                   :
DisplayGroup                  :
Group                         :
Enabled                       : True
Profile                       : Any
Platform                      : {}
Direction                     : Inbound
Action                        : Block
EdgeTraversalPolicy           : Block
LooseSourceMapping            : False
LocalOnlyMapping              : False
Owner                         :
PrimaryStatus                 : OK
Status                        : The rule was parsed successfully from the store. (65536)
EnforcementStatus             : NotApplicable
PolicyStoreSource             : PersistentStore
PolicyStoreSourceType         : Local
RemoteDynamicKeywordAddresses :
PolicyAppId                   :
```

PowerShell "Block" Configuration

Block SMB
Block RDP
Block ICMP

Inbound Rules

Name	Group	Profile	Enabled	Action	Override	Program	Local Address	Remote Address	Protocol	Local Port	Remote Port	Authorized Users	Authorized Compute
Allow ICMP from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	ICMPv4	Any	Any	Any	Any
Allow RDP from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	TCP	3389	Any	Any	Any
Allow SMB from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	TCP	445, 139	Any	Any	Any
Block ICMP from GuestNet		All	Yes	Block	No	Any	Any	10.10.20.0/24	ICMPv4	Any	Any	Any	Any
Block RDP from GuestNet		All	Yes	Block	No	Any	Any	10.10.20.0/24	TCP	3389	Any	Any	Any
Block SMB from GuestNet		All	Yes	Block	No	Any	Any	10.10.20.0/24	TCP	445, 139	Any	Any	Any

Screenshots Courtesy of Windows Powershell

4.3 Network Isolation (StaffNet vs GuestNet)

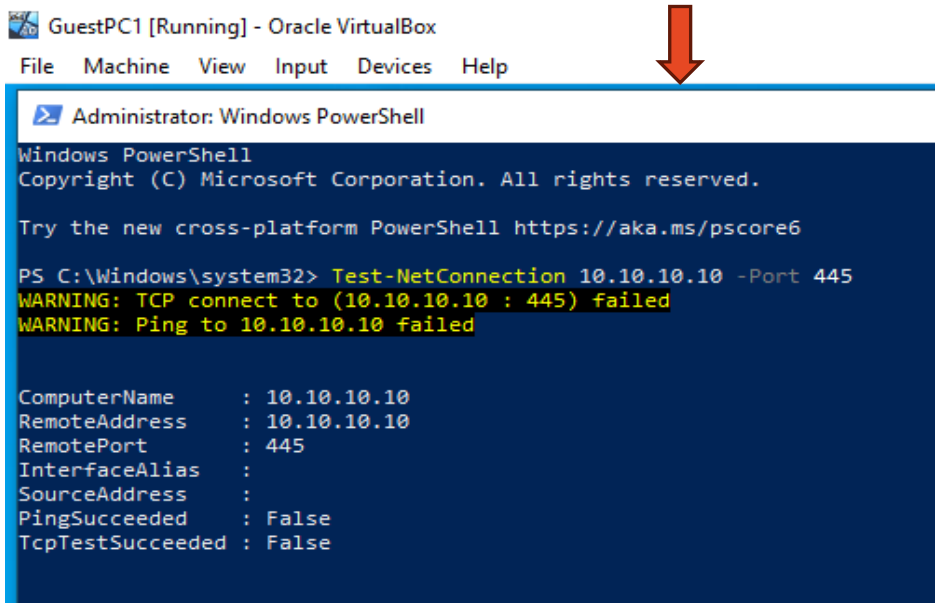
- Usage of internal network adapters to simulate VLAN isolation:
  How is it achieved? At the network layer, Staff machines are bound to the Host-only (vboxnet0) and Guest machines bound to Internet Network (GuestNet). These segments do not bridge to each other. (Refer to 4.2, point on network separation for illustration.

- Usage of Windows Firewall to prevent Guest-PC from reaching staff resources:
  How is it achieved? At the Host layer, File/Backup VMs are set to static IP within StaffNet. The same server/VM is then configured with an inbound rule that blocks TCP, FTP, SMB, RDP and ICMP communication from Guest-PC to access File/Backup, at a certain port. (Refer to 4.2, point on network separation for illustration.

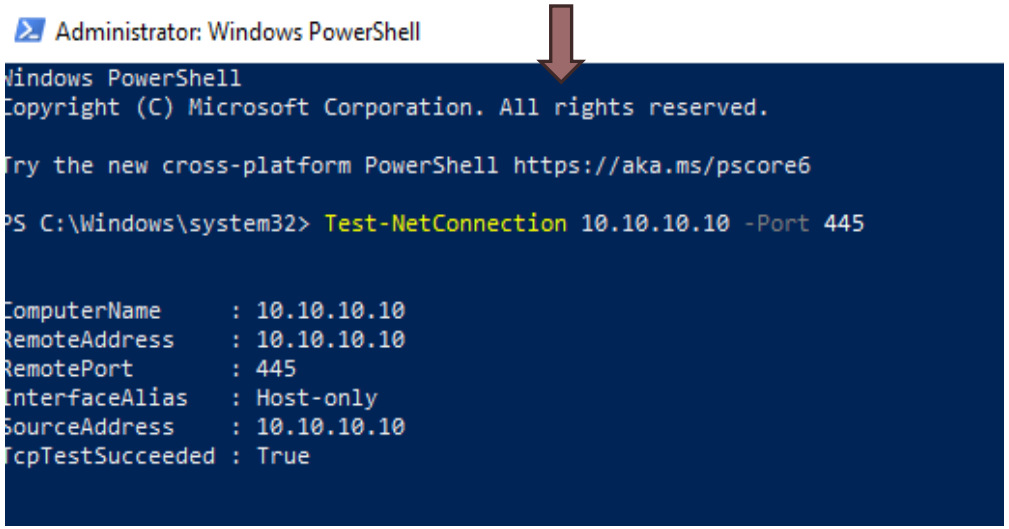
4.3 Network Isolation (StaffNet vs GuestNet)

- Demonstrate network segmentation with labelled screenshots:

*** Illustration (8) – GuestPC1 [10.10.20.11/24] trying to access FileBackupServer [10.10.10.10/24] and fails as expected**



*** Illustration (9) – StaffPC1Host [10.10.10.100/24] trying to access the FileBackupServer [10.10.10.10/24] and succeeds as expected**



Screenshots Courtesy of Windows Powershell

4.4 Data Backup and Printer Sharing Strategy

- File sharing via Windows Shares:
  How is it achieved? This can be achieved by means of creating a Shares Folder, then a staff group and allocating Folder access to this group. Or by removing “everyone” to the “Shares” (revoke access). *(Refer to point 4.6 and Annexure A 5.2.3)*

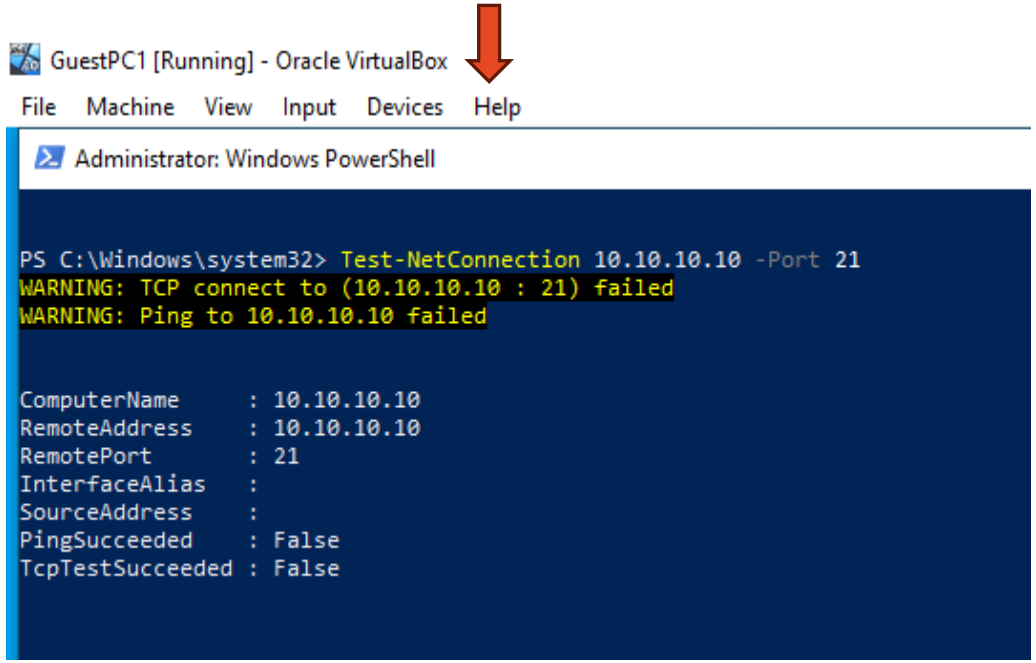
- Scheduling Backups:
  How is it achieved? This can be achieved by downloading Veeam Agent for Windows, updating the Windows OS, create a disk recovery job, configuring backup job on files based on the backup policy outline in the following bullet point. *(Refer to point 4.6 and Annexure A 5.2.4)*

- Backup Retention Policy:
  How is it achieved? Windows option (simple): Staff PCs and FileBackupServer VM to protect data. *(Refer to point 4.6)*
  - **Target:** \\files\Backups
  - **Schedule:** Daily incremental **19:00**; Weekly full **Sun 20:00**
  - **Retention: 30 daily / 4 weekly / 3 monthly**

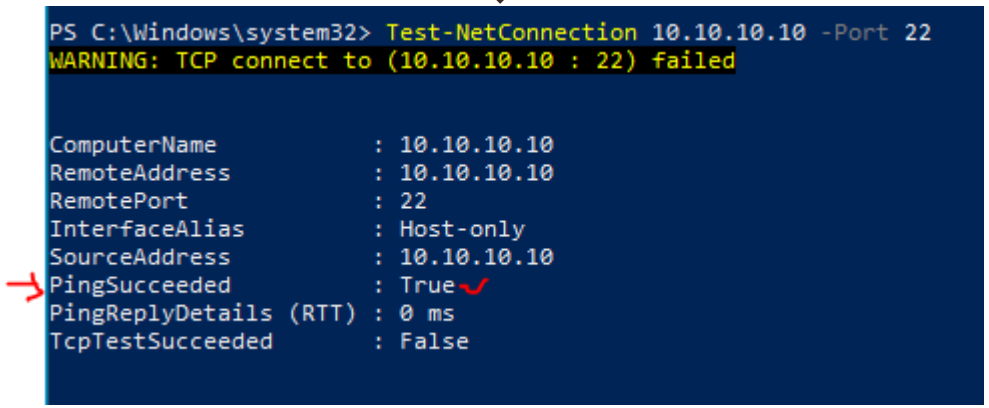
4.5 Windows Firewall Security Rules

- Ability to block/allow specific ports *(Refer to Annexure 5.2)*:

*** Illustration (10) – Telnet FileBackupServer on Port 21 to ensure that FTP cannot be achieved from GuestPC1**



*** Illustration (11) – Telnet FileBackupServer on Port 22 to ensure that SSH can be achieved from StaffPC1**

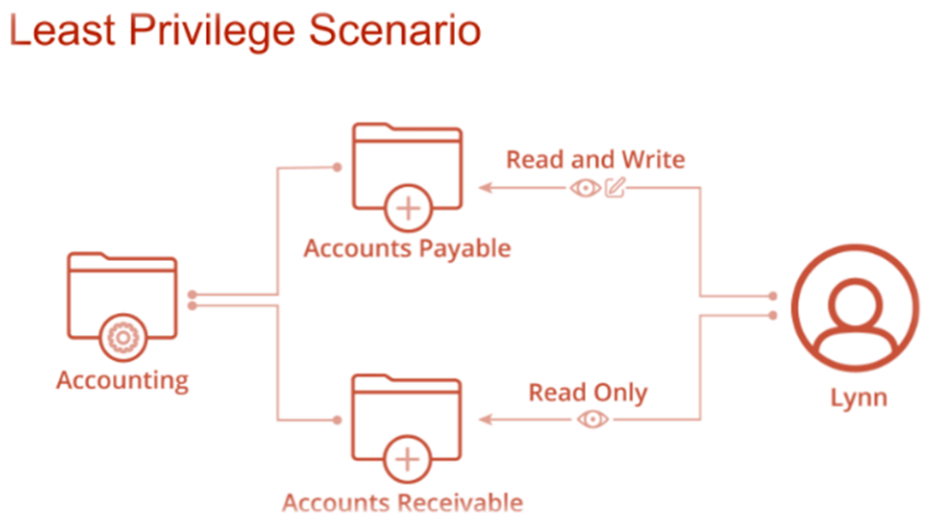


Screenshots Courtesy of Windows Powershell

4.5 Windows Firewall Security Rules (Ctd.)

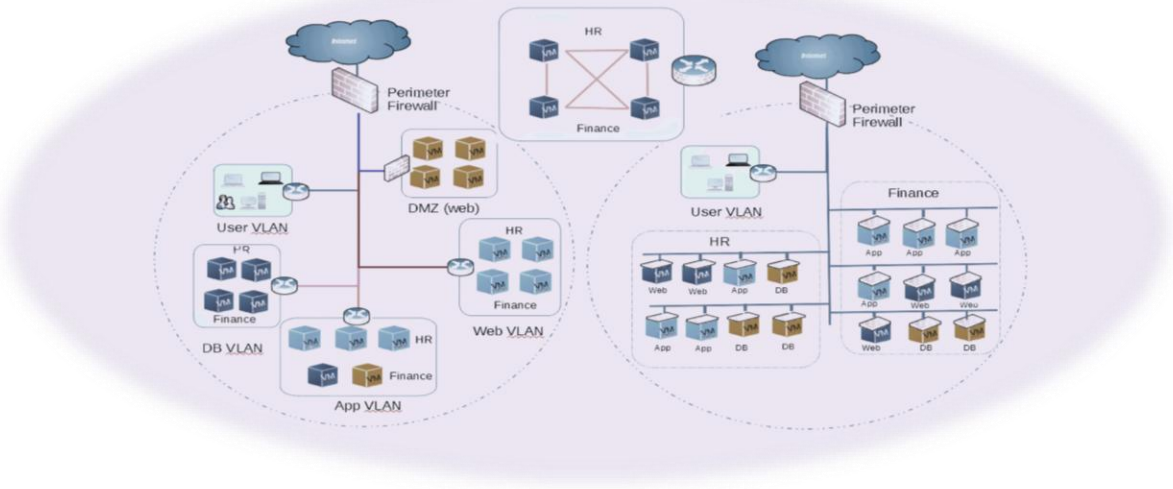
- Firewall rules using industry best practices:
  - **Least privilege (PoLP):** Only required ports are allowed, and only from trusted subnet.
  - **Segmentation/Zero-Trust:** Guest network is untrusted—no east-west access.
  - **Auditability:** Named rules with scope make reviews and audits straightforward.

*** Illustration (12) – PoLP (Link on Annexure 5.2)**



Screenshot Courtesy of Windows Publisher

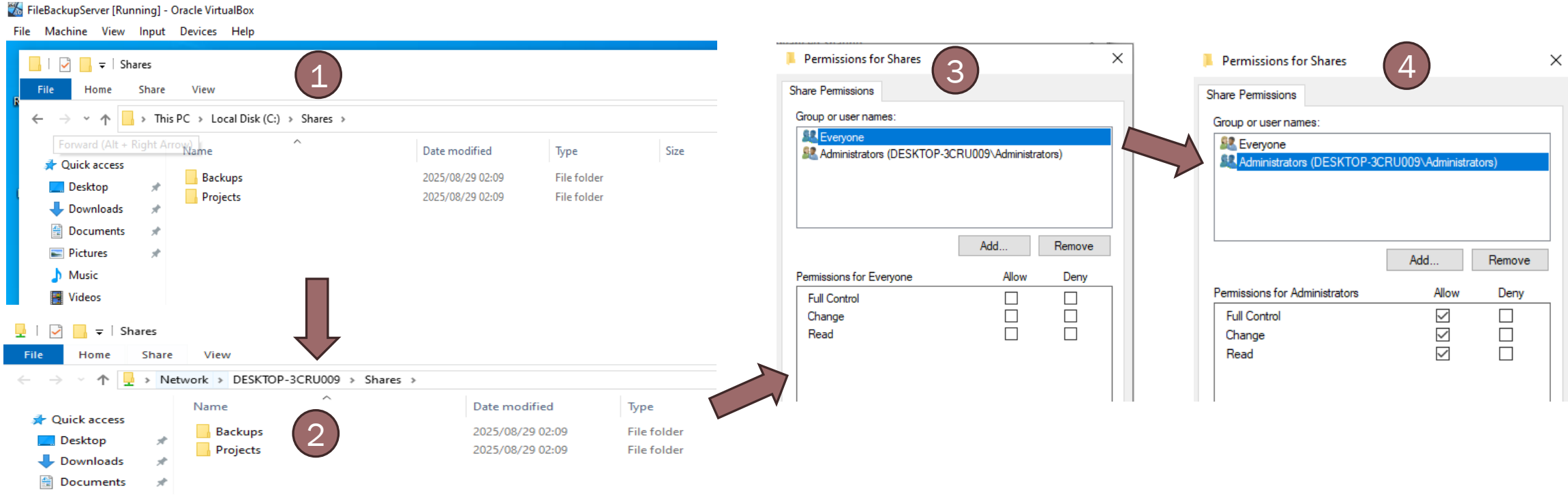
*** Illustration (13) – Zero-Trust (Link on Annexure 5.2)**



Screenshot Courtesy of Windows Publisher

4.6 Demonstration of Core Functionalities

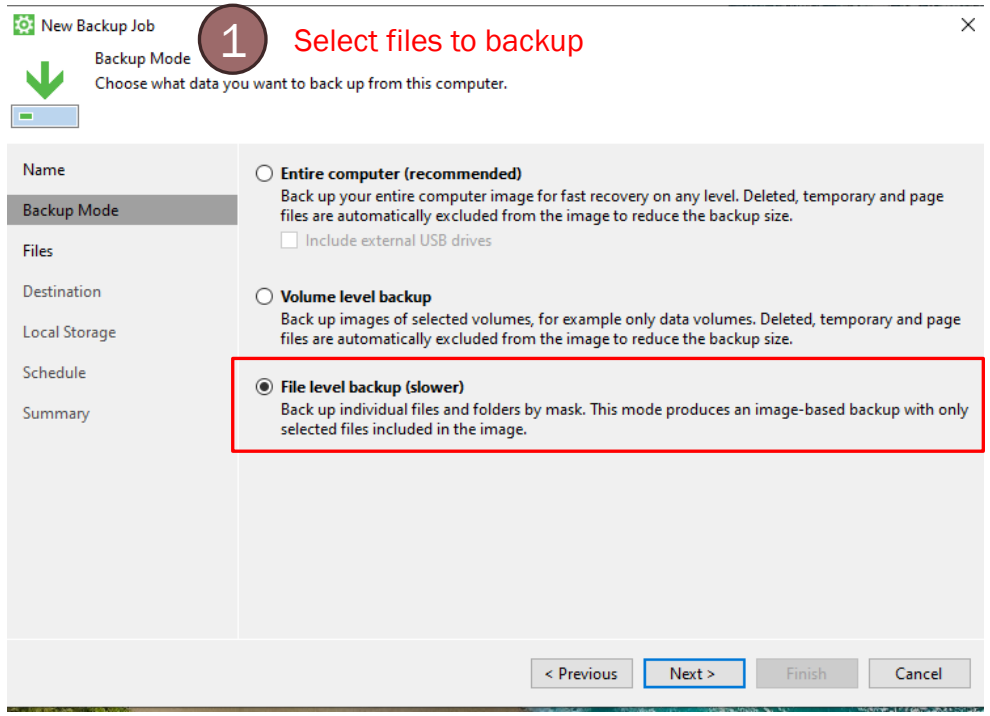
- **File sharing between staff PCs:** (0) StaffGroup Created, (1) File Shares Created, (2) Enable Network Sharing, (3) Allocate access to Admin and StaffGroup, (4) Revoke “Everyone”

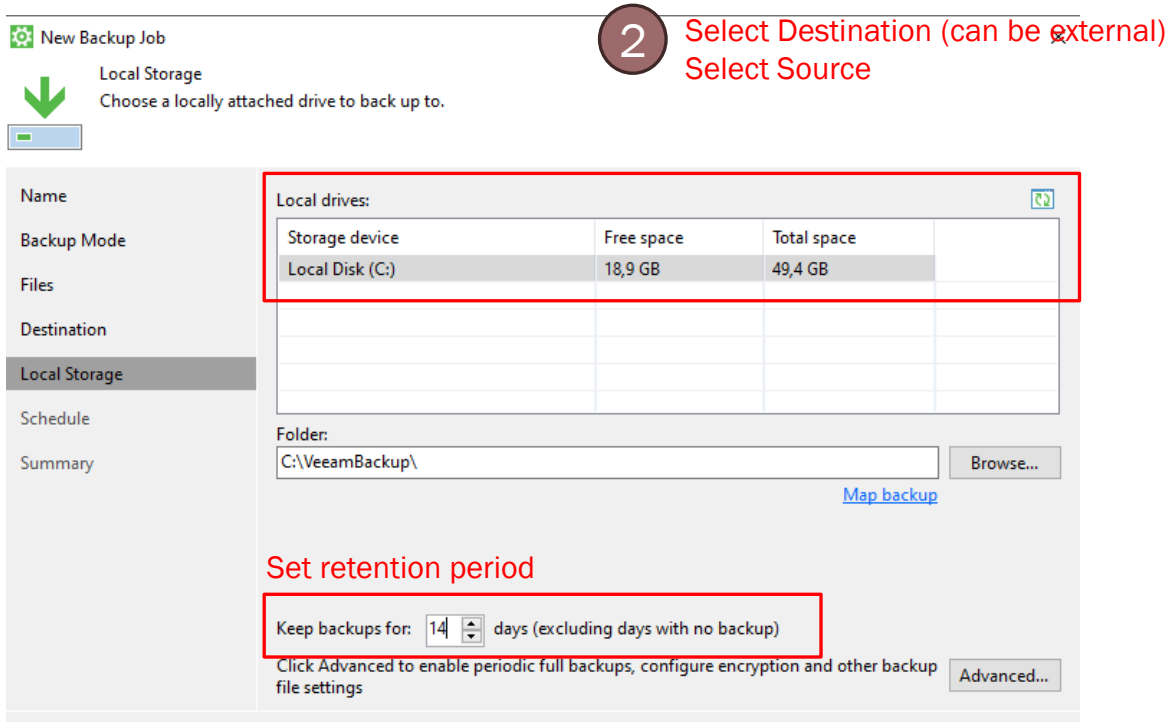


Screenshot Courtesy of Windows Server

4.6 Demonstration of Core Functionalities (Ctd.)

- **Scheduled Backups:** (-1) Download and install Veeam Agent for windows, (0) Create recovery for OS, (1) Create Scheduled Backup Job

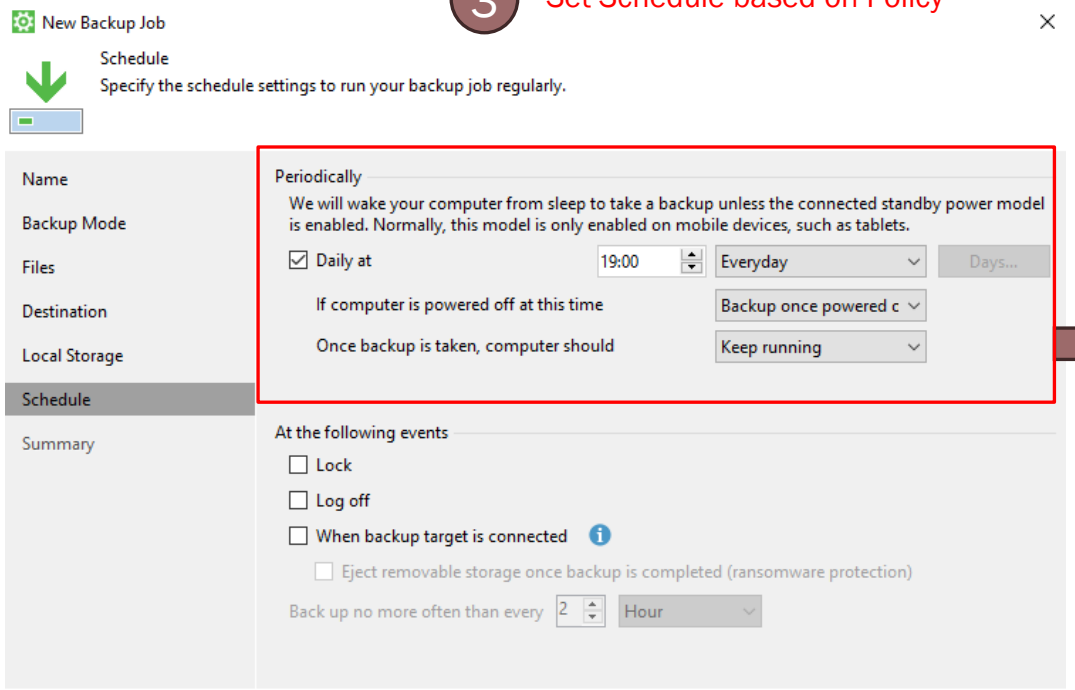




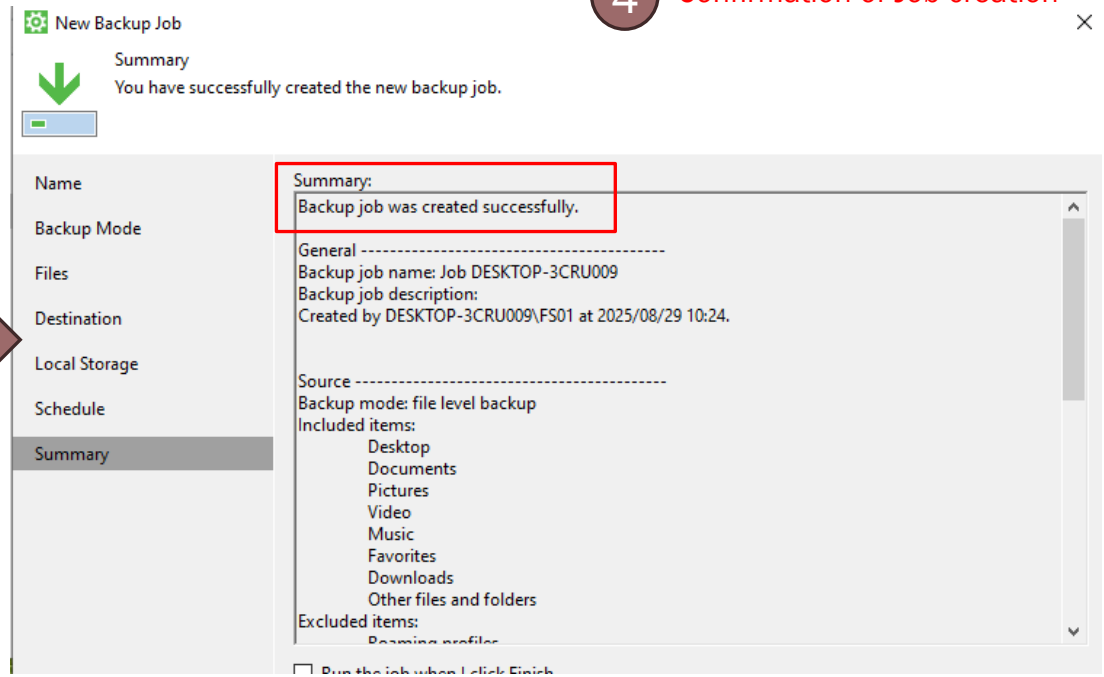
4.6 Demonstration of Core Functionalities (Ctd.)

- Scheduled Backups (ctd.)

3 Set Schedule based on Policy



4 Confirmation of Job creation



4.6 Demonstration of Core Functionalities (Ctd.)

- Description of how features were tested:
  - (1) VirtualBox Test: Download latest VirtualBox for Windows, Download Linux 2.6, Downloand Windows 10 ISO, setup and configure VMs.
  - (2) Backup Test: Download Veeam Agent, Create backup job, Run job and check “Last result: Success” with timestamp.

4.6 Demonstration of Core Functionalities (Ctd.)

(3) Network Test:

- Windows: ifconfig, netsh, Test-NetConnection {IP} –Port {XX}
- Linux: ipa, smbclient –L //{IP} –U {username}

Machine	Command / UI	Expectation
File-Backup	ipconfig, **NAT = DHCP, StaffNet = 10.10.10.10/24; no GW/DNS**	Default route only via NAT
Staff-PC2	ip a / ip route	enp0s8=10.10.10.12/24, no default via enp0s8; default via NAT device
Guest-PC	ipconfig, route print	GuestNet=10.10.20.11/24 without GW; default via NAT
Staff-PC1	Host-Only adapter GUI (vboxnet0)	10.10.10.100/24, no gateway
Any VM	Test-NetConnection 8.8.8.8 or ping	Internet works via NAT
Staff ↔ File-Backup	ping 10.10.10.10 / \\10.10.10.10\Projects	Reachable (firewall allows StaffNet later)
Guest → File-Backup	Test-NetConnection 10.10.10.10 -Port 445	**Will fail** after firewall rules in 2.2

5. Annexure

5.1. Annexure on Point 3

5.2. Annexure on Point 4

5.3. Additional Considerations

5.1 Annexure on 3

- *(A 5.1.1) One Pager Visual*

Greenleaf Architects – SOHO Network One-Pager

Topology • Firewall • Backups • Test Plan

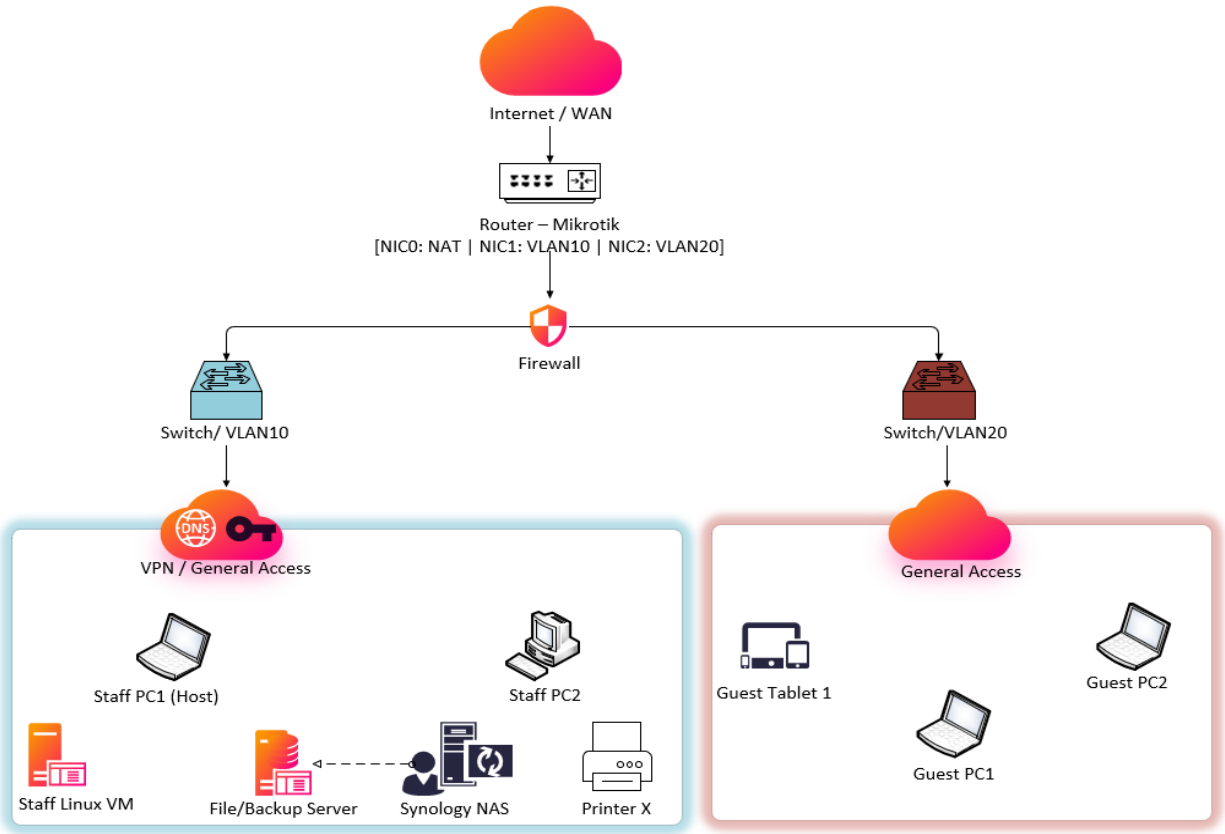
Topology (IPs)
Router (NAT) ⇄ StaffNet 10.10.10.0/24 ⇄ {File/Backup 10.10.10.10, NAS 10.10.10.5, Staff-PC1 10.10.10.11, Staff-PC2 10.10.10.12} Router (NAT) ⇄ GuestNet 10.10.20.0/24 ⇄ {Guest-PC 10.10.20.11}

Firewall policy (Windows Firewall)
ALLOW StaffNet → SMB(445), RDP(3389), ICMP to File/Backup + NAS. BLOCK GuestNet → SMB(445), RDP(3389), ICMP to File/Backup + NAS. Default Outbound: Allow; Inbound: Scoped to subnets.

File services & backups
Shares: \\files\Projects, \\files\Backups (Staff only). Backups: Daily incremental 19:00; Weekly full Sun; Retention 30d/4w/3m. NAS: RAID1 (2×4TB), snapshots; optional weekly USB/offsite copy.

Test checklist
1) Staff-PC1 can map \\files\Projects; Guest-PC denied. 2) Ping/RDP File/Backup from Staff allowed; from Guest blocked. 3) NAS shared folder accessible from Staff only. 4) Backup job runs; verify latest restore point.

Notes
Create VirtualBox internal nets: StaffNet, GuestNet; add NAT for internet. Use snapshots before firewall changes; document rule screenshots.

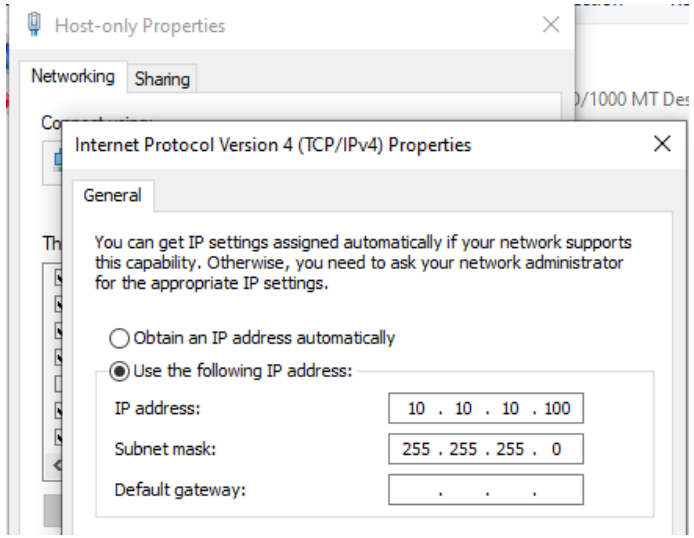


Screenshot Courtesy of Windows Publisher

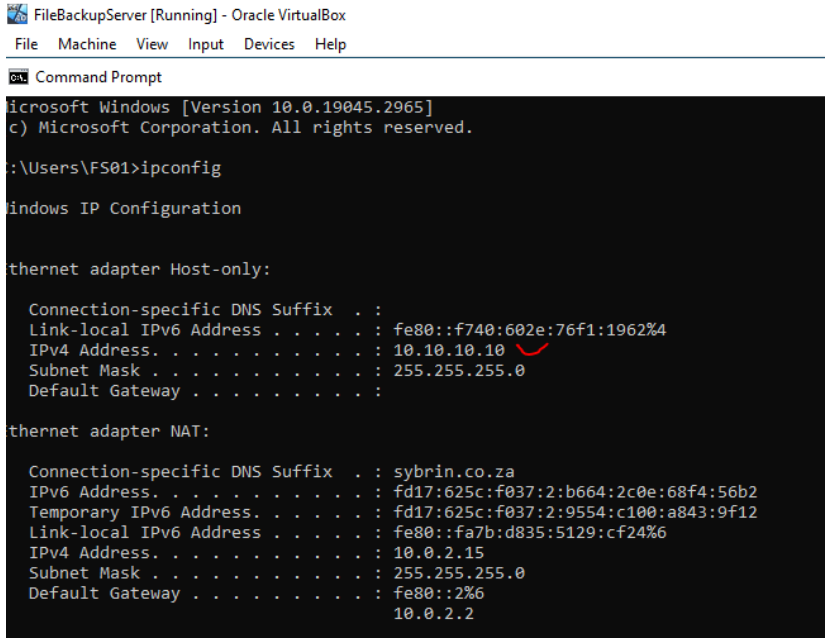
5.2 Annexure on 4

- *(A 5.2.1) IP Config*

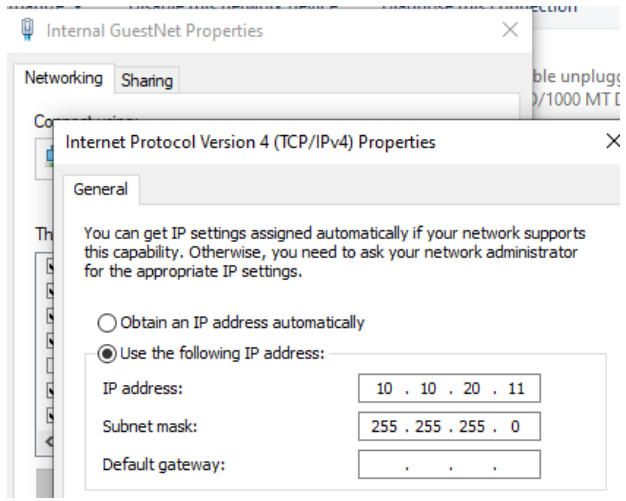
Staff-PC1 Host – IP 10.10.10.100



Staff-PC1 Host – IP 10.10.10.10



Guest-PC1 – IP 10.10.20.11



Screenshots from Windows

5.2 Annexure on 4

- *(A 5.2.2) Firewall Rule Settings*

Inbound Rules

Name	Group	Profile	Enabled	Action	Override	Program	Local Address	Remote Address	Protocol	Local Port	Remote Port	Authorized Users	Authorized Computer
Allow ICMP from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	ICMPv4	Any	Any	Any	Any
Allow RDP from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	TCP	3389	Any	Any	Any
Allow SMB from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	TCP	445, 139	Any	Any	Any
Allow SSH from StaffNet		All	Yes	Allow	No	Any	Any	10.10.10.0/24	TCP	22	Any	Any	Any
Block FTP from GuestNet		All	Yes	Block	No	Any	Any	10.10.20.0/24	TCP	21	Any	Any	Any
Block ICMP from GuestNet		All	Yes	Block	No	Any	Any	10.10.20.0/24	ICMPv4	Any	Any	Any	Any
Block RDP from GuestNet		All	Yes	Block	No	Any	Any	10.10.20.0/24	TCP	3389	Any	Any	Any
Block SMB from GuestNet		All	Yes	Block	No	Any	Any	10.10.20.0/24	TCP	445, 139	Any	Any	Any

SSH Detailed Configuration

Allow SSH from StaffNet Properties

General | Programs and Services | Remote Computers
Protocols and Ports | Scope | Advanced | Local Principals | Remote Users

Protocols and ports

Protocol type: TCP
Protocol number: 6

Local port: Specific Ports
22
Example: 80, 443, 5000-5010

Remote port: All Ports
Example: 80, 443, 5000-5010

Internet Control Message Protocol (ICMP) settings: Customize...

FTP Detailed Configuration

Block FTP from GuestNet Properties

General | Programs and Services | Remote Computers
Protocols and Ports | Scope | Advanced | Local Principals | Remote Users

Protocols and ports

Protocol type: TCP
Protocol number: 6

Local port: Specific Ports
21
Example: 80, 443, 5000-5010

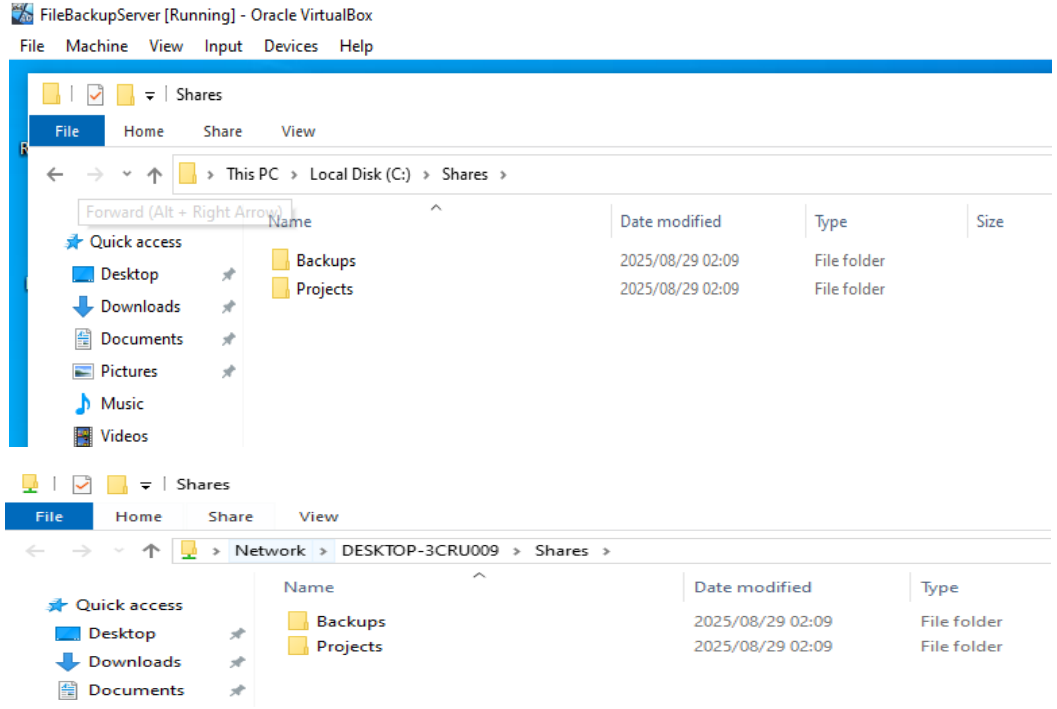
Remote port: All Ports
Example: 80, 443, 5000-5010

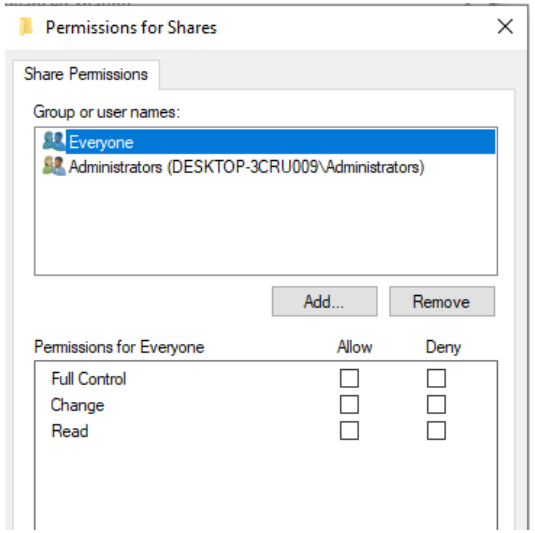
Internet Control Message Protocol (ICMP) settings: Customize...

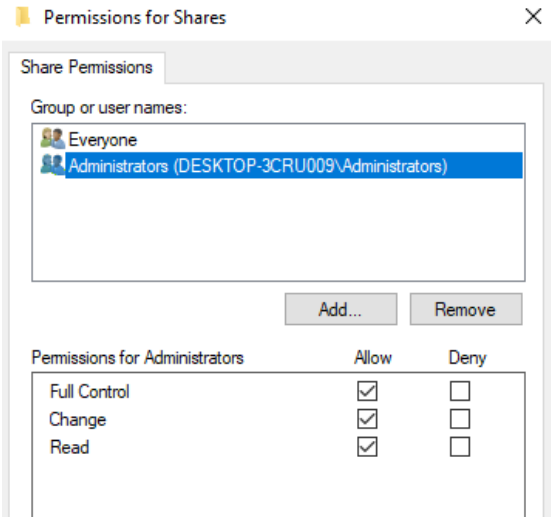
Screenshots Courtesy of Windows Firewall

5.2 Annexure on 4

- *(A 5.2.3) File Sharing*





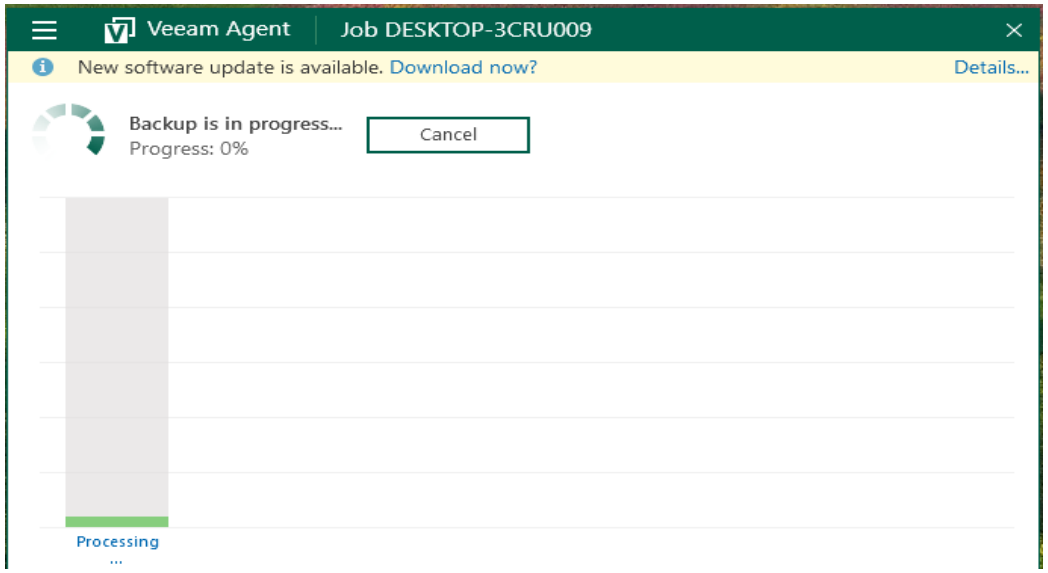


Screenshots Courtesy of Windows Server

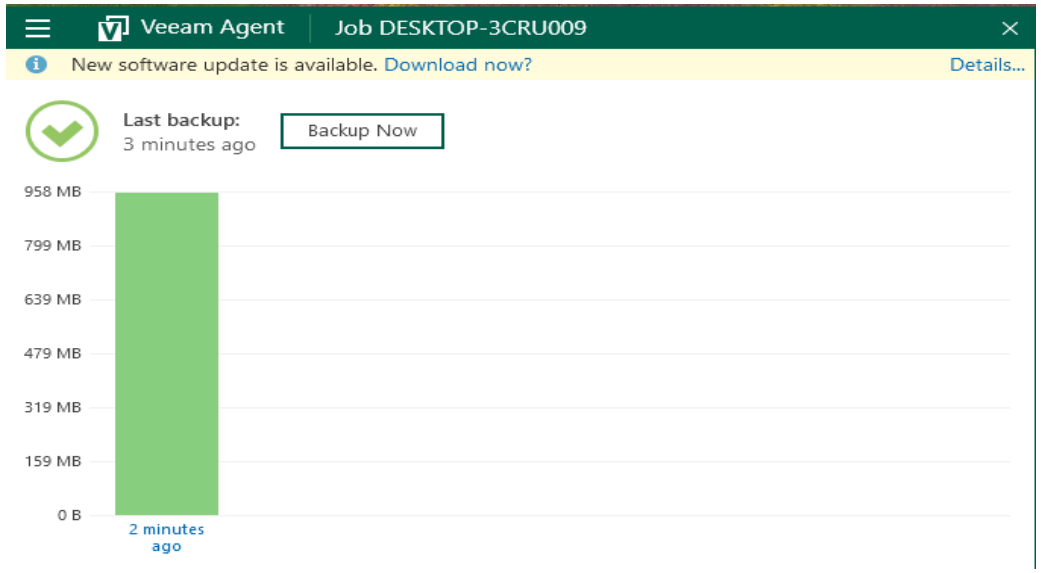
5.2 Annexure on 4

- *(A 5.2.4) Backup Status Confirmation*

Backup in Progress



Backup Completion



Screenshots Courtesy of Veam

5.2 Annexure on 4

- *(A5.2.5) Firewall Best Practices Links*

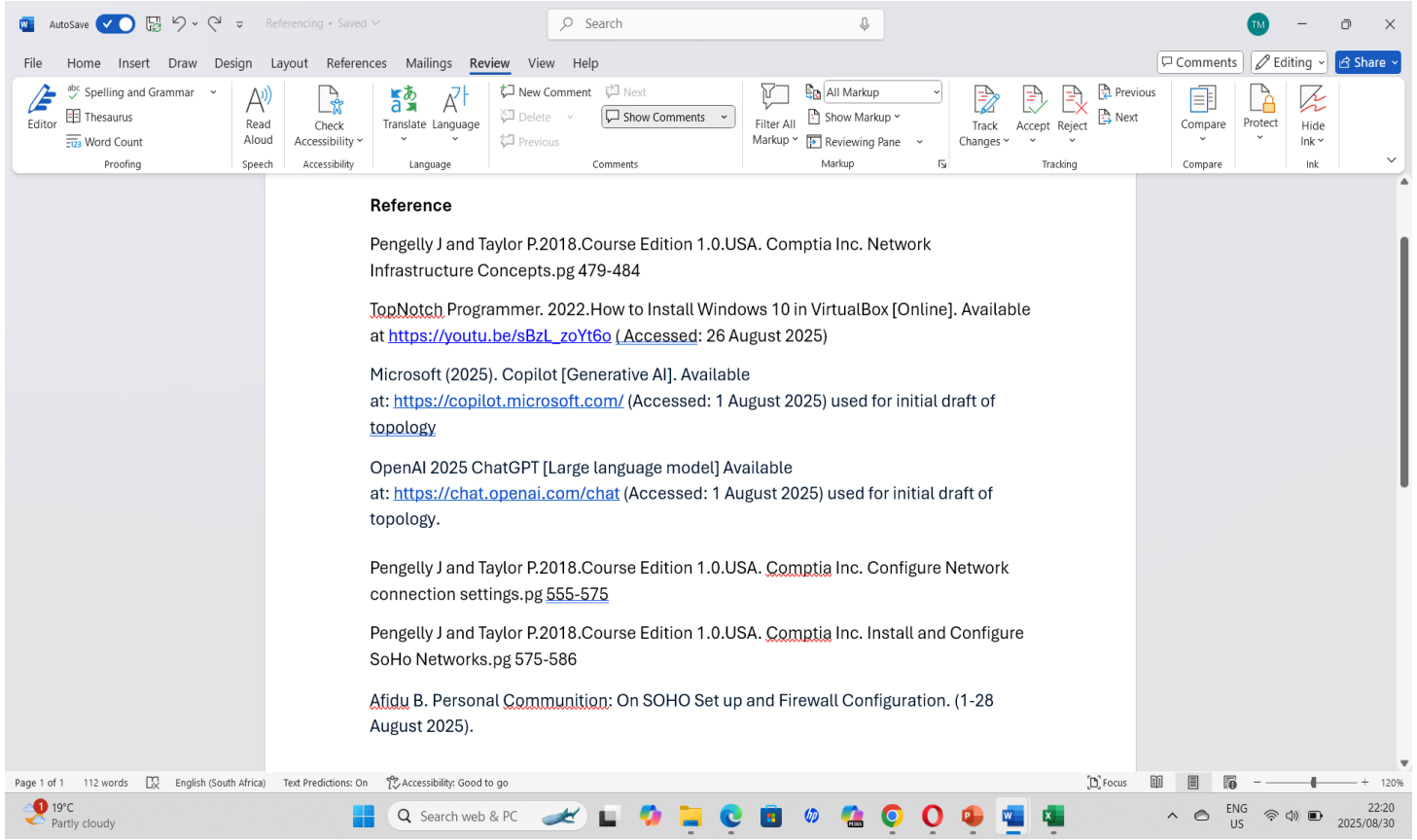
Zero-Trust: https://thesecmaster.com/blog/5-step-strategy-to-implement-micros-segmentations

PoLP: https://s3.tebi.io/7504.eu.org/Cybersecurity/Part%2003-Module%2001-Lesson%2002_Defending%20Computer%20Systems%20and%20Security%20Principles/12.%20Principle%20of%20Least%20Privileged.html

5.3 Additional Considerations

Risk	Impact	Mitigation
Misconfigured firewall rules	Service exposure or lockout	Standard templates, two-person review, pre-change snapshot, rollback plan
Single NAS/drive failure (Standard Topology)	Data loss / downtime	RAID1, SMART monitoring, snapshots, weekly external copy, restore drills
	Performance degradation	Capacity checks, staged backups, right-sizing, growth plan

5.5 Acknowledgements



Reference

Pengelly J and Taylor P.2018.Course Edition 1.0.USA. Comptia Inc. Network Infrastructure Concepts.pg 479-484

TopNotch Programmer. 2022.How to Install Windows 10 in VirtualBox [Online]. Available at https://youtu.be/sBzL_zoYt6o (Accessed: 26 August 2025)

Microsoft (2025). Copilot [Generative AI]. Available at: https://copilot.microsoft.com/ (Accessed: 1 August 2025) used for initial draft of topology

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Pengelly J and Taylor P.2018.Course Edition 1.0.USA. Comptia Inc. Configure Network connection settings.pg 555-575

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Afidu B. Personal Communition: On SOHO Set up and Firewall Configuration. (1-28 August 2025).

Screenshot Courtesy of Microsoft Word